

Samedh Sachin Kari

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EDUCATION

University of Pennsylvania Master of Science in Engineering, Electrical Engineering Relevant Coursework: Estimation and Detection Theory, Graph Neural Networks, Optimization Theory	Philadelphia, PA May 2028
Indian Institute of Space Science & Technology Bachelor of Technology, Electronics & Communication Engineer (Avionics) Thesis: Pose Estimation using Flash LiDAR	Thiruvananthapuram, India Jun 2025 GPA: 3.29/4

EXPERIENCE

Space Applications Centre, Indian Space Research Organisation Scientist / Engineer - 'SC' (Full-time), Sensor Development Area	Ahmedabad, India Jul 2025 - Jul 2026
- Developed target-adaptive LiDAR scanning strategies for autonomous docking applications, reducing scan time by 97%. - Designed a novel state-estimation filter for LiDAR-based 6-DOF spacecraft pose tracking, enabling relative navigation with 1 mm position accuracy and 1° attitude accuracy at 100 m operating range. - Implemented real-time LiDAR scanning, pose estimation and filtering software in Python and C++.	
U. R. Rao Satellite Centre, Indian Space Research Organisation Research Project Intern, Control Dynamics Design Group	Bengaluru, India Jan 2025 - Apr 2025
- Developed an ICP-based pose estimation algorithm for target-chaser systems, determining 6-DOF position and orientation from dense point cloud inputs captured by spacegrade LiDAR systems. - Applied Kalman filtering to refine pose and velocity estimates, improving stability and noise resilience of the system, and achieved accuracy of 5 mm for position and 3° for orientation. - Integrated the final pose and motion estimates as closed-loop control inputs within a target-chaser simulation, implementing the full pipeline in C++.	
U. R. Rao Satellite Centre, Indian Space Research Organisation Summer Research Intern, Control Dynamics Design Group	Bengaluru, India May 2024 - Jul 2024
- Developed a comprehensive mathematical model for every component of Flash LiDAR system. - Implemented the model as a Finite-Difference Time-Domain (FDTD) simulation in C for analysis. - Created an original LiDAR data clustering and preprocessing algorithm that discards about 65% noisy data early, accelerating subsequent computations.	

PUBLICATIONS

Samedh Sachin Kari, A. Arockia Bazil Raj and Balasubramanian. K, *Evolutionary Developments of Today's Remote Sensing Radar Technology - Right From the Telemobiloscope: A review*, **IEEE Geoscience and Remote Sensing Magazine**, March 2024.

SCHOLASTIC ACHIEVEMENTS

- All India Rank **452** in GATE 2024 (ECE) out of 63,092 candidates (**99.28** percentile)
- Top **0.6%** in 1.3 million candidates overall with **99.99** subject percentile for mathematics in JEE 2021
- Regional Mathematics Olympiad (RMO) Awardee 2018 (Top 30 in Maharashtra region and top 300 among 10,000+ candidates across India)
- Gold Honour in International Youth Math Challenge (**Top 1 %** international participants)

TECHNICAL SKILLS

Signal Processing & Controls	Pose Estimation and State Estimation, Mathematical Modelling, FDTD Modelling, Kalman Filtering, PID Control
Programming	Python, C, C++, MATLAB and Simulink, Verilog
Tools	Linux, L ^A T _E X, Git, Microsoft Office